Q1. JEE Main 2026 (28 January Shift 2)

Given below are two statements: **Statement I:** The function $f : \mathbf{R} \to \mathbf{R}$ defined by $f(x) = \frac{x}{1+|x|}$ is one-one.
Statement II: The function $f : \mathbf{R} \to \mathbf{R}$ defined by $f(x) = \frac{x^2+4x-30}{x^2-8x+18}$ is many-one.
In the light of the above statements, choose the correct answer from the options given below :
(1) Both Statement I and Statement II are true
(2) Statement I is true but Statement II is false
(3) Statement I is false but Statement II is true
(4) Both Statement I and Statement II are false

Q2. JEE Main 2026 (28 January Shift 2)

The sum of all the elements in the range of $f(x) = \text{Sgn}(\sin x) + \text{Sgn}(\cos x) + \text{Sgn}(\tan x) + \text{Sgn}(\cot x)$, $x \neq \frac{\text{n}\pi}{2}, \text{n} \in \mathbf{Z}$, where $\text{Sgn}(\text{t}) = \begin{cases} 1, \text{ if t} > 0 \\ -1, \text{ if t} < 0 \end{cases}$, is :

(1) 2 (2) 0 (3) 4 (4) -2

Q3. JEE Main 2026 (24 January Shift 2)

Let f be a function such that $3f(x) + 2f\left(\frac{m}{19x}\right) = 5x, x \neq 0$, where $m = \sum_{i=1}^{9}(i)^2$. Then $f(5) - f(2)$ is equal to

(1) 18 (2) 36 (3) 9 (4) -9

Q4. JEE Main 2026 (22 January Shift 2)

Let the domain of the function $f(x) = \log_3 \log_5 \left(7 - \log_2\left(x^2 - 10x + 85\right)\right) + \sin^{-1}\left(\left|\frac{3x-7}{17-x}\right|\right)$ be $(\alpha, \beta]$. Then $\alpha + \beta$ is equal to :

(1) 9 (2) 10 (3) 12 (4) 8

Q5. JEE Main 2026 (22 January Shift 2)

Let f and g be functions satisfying $f(x+y) = f(x)f(y), f(1) = 7$ and $g(x+y) = g(xy), g(1) = 1$, for all $x, y \in \mathbf{N}$. If $\sum_{x=1}^{\text{n}}\left(\frac{f(x)}{\text{g}(x)}\right) = 19607$, then n is equal to:

(1) 7 (2) 6 (3) 4 (4) 5

Q6. JEE Main 2026 (22 January Shift 1)

If the domain of the function $f(x) = \sin^{-1}\left(\frac{5-x}{3+2x}\right) + \frac{1}{\log_e(10-x)}$ is $(-\infty, \alpha] \cup [\beta, \gamma) - \{\delta\}$, then $6(\alpha + \beta + \gamma + \delta)$ is equal to

(1) 67 (2) 66 (3) 70 (4) 68

Q7. JEE Main 2026 (22 January Shift 1)

The number of distinct real solutions of the equation $x|x+4| + 3|x+2| + 10 = 0$ is

(1) 2 (2) 1 (3) 3 (4) 0

Q8. JEE Main 2026 (23 January Shift 2)

Consider two sets $A = \{x \in Z : |(|x-3|-3)| \leq 1\}$ and $B = \left\{x \in \mathbb{R} - \{1,2\} : \frac{(x-2)(x-4)}{x-1}\log_e(|x-2|) = 0\right\}$. Then the number of onto functions $f : A \to B$ is equal to

(1) 32 (2) 79 (3) 62 (4) 81

Q9. JEE Main 2026 (28 January Shift 1)

If $g(x) = 3x^2 + 2x - 3$, $f(0) = -3$ and $4g(f(x)) = 3x^2 - 32x + 72$, then $f(g(2))$ is equal to:

(1) $-\frac{7}{2}$ (2) $-\frac{25}{6}$ (3) $\frac{7}{2}$ (4) $\frac{25}{6}$

Q10. JEE Main 2026 (24 January Shift 1)

If the domain of the function $f(x) = \log_{(10x^2-17x+7)}\left(18x^2 - 11x + 1\right)$ is $(-\infty, a) \cup (b, c) \cup (d, \infty) - \{e\}$, then $90(a+b+c+d+e)$ equals:

(1) 170 (2) 307 (3) 316 (4) 177

ANSWERS AND SOLUTIONS

1. (1)	**2.** (1)	**3.** (1)	**4.** (1)	**5.** (4)	**6.** (3)	**7.** (2)	**8.** (3)
9. (3)	**10.** (3)						

1. **(1)** Statement I: $f(x) = \frac{x}{1+|x|}$. For $x \geq 0$, $f(x) = \frac{x}{1+x}$ is strictly increasing. For $x < 0$, $f(x) = \frac{x}{1-x}$ with $f'(x) = \frac{1}{(1-x)^2} > 0$, also strictly increasing. The function is continuous at $x = 0$, so f is strictly increasing on $\mathbb{R}$ and hence one-one. Statement I is true.

Statement II: $f(x) = \frac{x^2+4x-30}{x^2-8x+18}$. The denominator $(x-4)^2 + 2 > 0$ for all x.

$f'(x) = \frac{-12(x^2-8x+14)}{(x^2-8x+18)^2}$

$f'(x) = 0 \Rightarrow x = 4 \pm \sqrt{2}$. Since f' changes sign, f is not monotonic and hence many-one. Statement II is true.

Both statements are true.

2. **(1)** $x \in (0, \pi/2) \Rightarrow y = 1 + 1 + 1 + 1 = 4$

$x \in (\pi/2, \pi) \Rightarrow y = 1 - 1 - 1 - 1 = -2$

$x \in (\pi, 3\pi/2) \Rightarrow y = -1 - 1 + 1 + 1 = 0$

$x \in (3\pi/2, 2\pi) \Rightarrow y = -1 + 1 - 1 - 1 = -2$

$\therefore$ Range of y is $\{-2, 0, 4\}$

Required sum $= -2 + 0 + 4 = 2$

3. **(1)** $m = \sum_{i=1}^{9} i^2 = \frac{9 \cdot 10 \cdot 19}{6} = 285$, so $\frac{m}{19} = 15$.

The equation becomes $3f(x) + 2f\left(\frac{15}{x}\right) = 5x$ ...(i)

Replace x by $\frac{15}{x}$: $3f\left(\frac{15}{x}\right) + 2f(x) = \frac{75}{x}$ ...(ii)

From $3 \times (i) - 2 \times (ii)$: $5f(x) = 15x - \frac{150}{x}$, so $f(x) = 3x - \frac{30}{x}$.

$f(5) = 15 - 6 = 9$, $f(2) = 6 - 15 = -9$.

$f(5) - f(2) = 9 - (-9) = 18$.

4. **(1)** For $\log_3 \log_5(\cdot)$ we need $\log_5(\cdot) > 0$

So $7 - \log_2(x^2 - 10x + 85) > 1$, giving $x^2 - 10x + 85 < 64$

$(x-3)(x-7) < 0$

$\Rightarrow 3 < x < 7$.

For $\sin^{-1}\left(\left|\frac{3x-7}{17-x}\right|\right)$ we need $|3x - 7| \leq |17 - x|$

i.e, $(x-6)(x+5) \leq 0 \Rightarrow -5 \leq x \leq 6$.

Intersection: $(3, 6]$, so $\alpha = 3, \beta = 6$.

$\alpha + \beta = 9$

5. **(4)** $f(x+y) = f(x)f(y)$ with $f(1) = 7$ gives $f(x) = 7^x$.

For g: put $y = 1$ in $g(x+y) = g(xy)$ to get $g(x+1) = g(x)$.

So $g(x) = g(1) = 1$ for all $x \in \mathbb{N}$.

$\sum_{x=1}^{n} \frac{f(x)}{g(x)} = \sum_{x=1}^{n} 7^x = \frac{7(7^n-1)}{6} = 19607$

$7^{n+1} - 7 = 117642$

$\Rightarrow 7^{n+1} = 117649 = 7^6$

$n + 1 = 6 \Rightarrow n = 5$

6. **(3)** $-1 \leq \dfrac{5-x}{2x+3} \leq$ & $10 - x > 0, 10 - x \neq 1$

$\left|\dfrac{5-x}{2x+3}\right| \geq 1$ & $x < 10$ & $x \neq 9$

$(5-x)^2 - (2x+3)^2 \leq 0$ & $x < 10$ & $4x \neq 9$

$(x+8)(3x-2) \geq 0$ & $x < 10$ & $x \neq 9$

$\Rightarrow (-\infty, -8] \cup \left[\dfrac{2}{3}, 10\right) - \{9\}$

$\Rightarrow (\alpha + \beta + \gamma + \delta) = 6\left(-8 + \dfrac{2}{3} + 10 + 9\right)$

$= 70$

7. **(2)** Equation: $x|x+4| + 3|x+2| + 10 = 0$.

Critical points: $x = -4, -2$.

Case 1:

$x \geq -2$: $x(x+4) + 3(x+2) + 10 = 0$

$\Rightarrow x^2 + 7x + 16 = 0$.

Discriminant $= 49 - 64 = -15 < 0$. No real roots.

Case 2:

$-4 \leq x < -2$: $x(x+4) - 3(x+2) + 10 = 0$

$\Rightarrow x^2 + x + 4 = 0$.

Discriminant $= 1 - 16 = -15 < 0$. No real roots.

Case 3:

$x < -4$: $-x(x+4) - 3(x+2) + 10 = 0$

$\Rightarrow x^2 + 7x - 4 = 0$.

$x = \frac{-7 \pm \sqrt{65}}{2}$.

$x = \frac{-7+\sqrt{65}}{2} \approx 0.53$ (rejected, not in $x < -4$). $x = \frac{-7-\sqrt{65}}{2} \approx -7.53$ (valid).

Number of distinct real solutions $= 1$.

8. **(3)** For set A:

$|(|x-3| - 3)| \leq 1 \Rightarrow 2 \leq |x-3| \leq 4$.

If $x \geq 3$: $5 \leq x \leq 7 \Rightarrow x \in \{5, 6, 7\}$.

If $x < 3$: $-1 \leq x \leq 1 \Rightarrow x \in \{-1, 0, 1\}$.

So $A = \{-1, 0, 1, 5, 6, 7\}$, $|A| = 6$.

For set B:

$\frac{(x-2)(x-4)}{x-1}\log_e(|x-2|) = 0$ with $x \in \mathbb{R} - \{1, 2\}$.

Either $\frac{(x-2)(x-4)}{x-1} = 0 \Rightarrow x = 4$ (since $x \neq 1, 2$), or $\log_e |x-2| = 0 \Rightarrow |x-2| = 1 \Rightarrow x = 3$ (since $x \neq 1$).

So $B = \{3, 4\}$, $|B| = 2$.

Number of onto functions from A to $B = 2^6 - 2 = 62$.

9. **(3)** Let $f(x) = ax + b$. Then $4g(f(x)) = 12(ax+b)^2 + 8(ax+b) - 12$.

Comparing with $3x^2 - 32x + 72$:

x^2: $12a^2 = 3 \Rightarrow a = \pm\frac{1}{2}$

Constant: $12b^2 + 8b - 12 = 72 \Rightarrow 3b^2 + 2b - 21 = 0 \Rightarrow b = \frac{7}{3}$ or $b = -3$. Since $f(0) = -3$, $b = -3$.

x: $8a(3(-3)+1) = -32 \Rightarrow -8a = -4 \Rightarrow a = \frac{1}{2}$.

So $f(x) = \frac{x}{2} - 3$.

$g(2) = 3(4) + 4 - 3 = 13$.

$f(g(2)) = f(13) = \frac{13}{2} - 3 = \frac{7}{2}$.

10. **(3)** For $f(x) = \log_{(10x^2-17x+7)}(18x^2 - 11x + 1)$, we need:

Base > 0: $10x^2 - 17x + 7 > 0 \Rightarrow 10(x-1)(x-7/10) > 0 \Rightarrow x < 7/10$ or $x > 1$

Base $\neq 1$: $10x^2 - 17x + 6 = 0 \Rightarrow x = 1/2$ or $x = 6/5$

Argument > 0: $18x^2 - 11x + 1 > 0 \Rightarrow 18(x-1/2)(x-1/9) > 0 \Rightarrow x < 1/9$ or $x > 1/2$

Intersecting conditions 1 and 3: $(-\infty, 1/9) \cup (1/2, 7/10) \cup (1, \infty)$

Removing base $= 1$ points: $x = 1/2$ is already excluded; $x = 6/5 \in (1, \infty)$ must be removed.

Domain: $(-\infty, 1/9) \cup (1/2, 7/10) \cup (1, \infty) - \{6/5\}$

So $a = 1/9,\ b = 1/2,\ c = 7/10,\ d = 1,\ e = 6/5$.

$90(a+b+c+d+e) = 90\left(\frac{1}{9} + \frac{1}{2} + \frac{7}{10} + 1 + \frac{6}{5}\right) = 10 + 45 + 63 + 90 + 108 = 316$